

# **vLLM: Enabling faster inference and serving for LLMs**



Large language models (LLMs) have revolutionized AI applications, but their efficiency and cost of deployment remain significant challenges. vLLM is a powerful library designed to enhance LLM inference and serving by improving speed and memory efficiency. Originally developed at UC Berkeley's Sky Computing Lab, vLLM has grown into a community-driven project with contributors from academia and industry.

## What is vLLM?

vLLM is a Python library that optimizes the inference and serving of LLMs. It offers:

- **High-speed performance** with state-of-the-art serving throughput.
- **Efficient memory management** using PagedAttention, which minimizes memory waste.
- **Continuous batching** of requests for improved scalability.

It supports various popular LLMs, such as LLaMA, OPT, Mixtral, StableLM, and Falcon. Compared to Hugging Face's Transformers, vLLM delivers up to **24x higher throughput** without requiring changes to the model architecture.

## Key Features of vLLM

vLLM introduces multiple optimizations to improve LLM inference:

- **Fast execution with CUDA** for enhanced GPU performance.
- **Support for quantization techniques** such as GPTQ, AWQ, INT4, INT8, and FP8 to reduce memory consumption.
- **Integration with FlashAttention and FlashInfer** for better efficiency.
- **Speculative decoding** to speed up text generation.

- **Chunked prefill** for improved processing of large input sequences.

## PagedAttention: The Core of vLLM

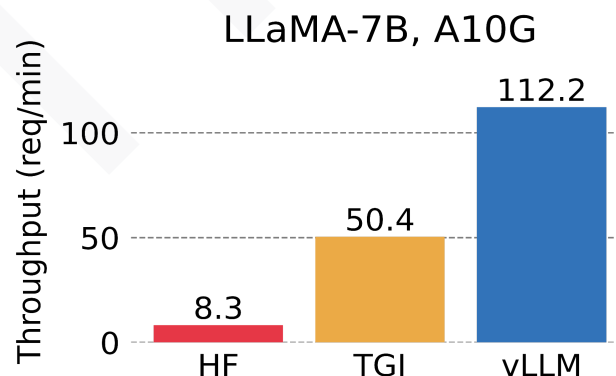
One of the biggest challenges in serving LLMs is memory management. Each input token generates attention keys and values (KV), which need to be stored for reference. This KV cache can take up massive memory, leading to inefficiencies in handling large-scale queries.

**PagedAttention solves this problem** by breaking down the KV cache into smaller, non-contiguous blocks, similar to how virtual memory works in operating systems. This reduces memory waste to less than **4%**, allowing more requests to be processed in parallel.

## How Fast is vLLM?

Performance benchmarks show that vLLM outperforms other inference solutions:

- **14x to 24x faster** than Hugging Face Transformers.
- **3.5x faster** than Hugging Face's Text Generation Inference (TGI), which was previously the fastest solution.
- Enables **5x more traffic** handling without requiring additional GPUs.



vLLM vs HuggingFace vs TGI throughput comparison

For businesses deploying AI-powered chatbots or virtual assistants, these improvements lead to reduced costs and better user experience.

vLLM vs HuggingFace inference time testing notebook [link](#).

### **vLLM vs. Ollama: A Quick Comparison**

- **vLLM**: Best for serving LLMs on **GPUs** with high throughput and parallel processing.
- **Ollama**: Designed for developers working with LLMs on **CPUs**, offering greater flexibility but lower performance.

### **Conclusion**

vLLM is a game-changer for LLM deployment, offering unmatched efficiency in speed and memory management. Whether you're running models locally or in the cloud, vLLM provides a scalable and cost-effective solution for optimizing AI applications.

Extra Resource:

1. [vLLM paper](#)
2. [Paper and Paged Attention Explanation](#)